INTERNATIONAL MONETARY FUND

RWANDA

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REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS[1]

Approved By
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The Debt Sustainability Analysis (DSA) was prepared jointly by the staffs of the International Monetary Fund and the International Development Association.

Risk of external debt distress	Moderate[2]
Overall risk of debt distress	Moderate
Granularity in the risk rating	Limited space to absorb shocks
Application of judgment	No
Macroeconomic projections	The DSA incorporates the expected impacts of the COVID-19 pandemic, with a sharp decline in real GDP growth (down by 6 percentage points relative to the pre-pandemic projection), exports of goods and services (40 percent decline), tax revenues (20 percent decline), and a significant widening of the fiscal deficit (up by about 6 percentage points of GDP).
Financing strategy	The DSA incorporates a proposed RCF disbursement equivalent to 50 percent of quota, together with prospective concessional financing from other development partners, including the World Bank's COVID-19 Emergency Response Project.
Realism tools flagged	No
Mechanical risk rating under the external DSA	Moderate
Mechanical risk rating under the public DSA	Moderate

[1] This debt sustainability analysis was conducted using the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries (LIC-DSF) that was approved in 2017. The fiscal year for Rwanda is from July–June; however, this DSA is prepared on a calendar year basis.

[2] Rwanda's debt-carrying capacity remains strong as its Composite Indicator (CI) score of 3.26 remains above the upper threshold of 3.05. This is based on the October 2019 WEO and CPIA 2018. There has been no change in debt coverage since the last DSA (July 2019, IMF CR 19/211), which covers the central government and state-owned enterprises (SOEs).

While Rwanda's debt remains sustainable, the risk of external and overall debt distress has moved to moderate from low relative to the previous DSA in April 2020 due to the impact of the global COVID-19 crisis. The baseline macroeconomic scenario reflects negative effects of the COVID-19 shock on growth, exports, and revenues, which sharply raise external and domestic financing needs in 2020.[3] This DSA assumes that the Rapid Credit Facility (RCF), debt service relief under the IMF's Catastrophe Containment and Relief Trust (CCRT), World Bank financial support,[4] and prospective concessional financing from other development partners fill the external financing gap created by the shock. The authorities have not yet requested debt service suspension from official bilateral creditors as envisaged under the Debt Service Suspension Initiative (DSSI), supported by the G-20 and Paris Club.[5] Compared to the previous DSA, the pandemic impact coupled with higher concessional loans from multilateral and bilateral donors and larger external borrowing related to an scaling up of the new international airport project is expected to entail a higher pace of accumulation of public and publicly guaranteed (PPG) external debt in the medium term, leading to a present value (PV) of total public debt peaking at 53 percent of GDP in 2025.[6]

The stress tests highlight that Rwanda is more susceptible to external shocks when compared to the pre-pandemic period even after the initial impact of the COVID-19 shock dissipates, and, as a result, indicate the moderate risk of debt distress (Figure 1 and 2).[7] Solvency indicators for the PV of external debt-to-GDP ratio and the PV of total public debt-to-GDP ratio are below the indicative threshold/benchmark under both the baseline scenario and the most extreme risk shock. However, the sharp decline in exports due to the COVID-19 shock makes one solvency indictor (the PV of external debt-to-export ratio) and one liquidity indicator (the external debt service-to-exports ratio) temporarily breach their relevant thresholds in 2020 under the baseline scenario. This risk is, however, mitigated by adequate reserves and external financing that had already been lined up. There are also multiple breaches to these indicators under the most extreme shock through heightened uncertainty surrounding exports. The servicing spike in external debt service in 2023 (due to the rollover of the 10-year Eurobond issued in 2013) causes a one-off breach to the other liquidity indicator of debt service-to-revenue ratio under the baseline scenario. This indicator also shows another one-off breach in 2028 under the most extreme shock, which is caused by the market financing shock halving the maturity of the 10-year Eurobond expected to be issued in 2023. In order to reduce the rollover risks, the authorities

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[3] The DSA incorporates the expected impacts of the COVID-19 pandemic, with a sharp decline in real GDP growth (down by 6 percentage points relative to the pre-pandemic projection), exports of goods and services (40 percent decline), tax revenues (20 percent decline), and a significant widening of the fiscal deficit (up by about 6 percentage points of GDP). Following the COVID-19 shock in 2020, it assumes economic recovery from 2021, with the economy gradually reverting to its pre-pandemic trend in the medium term.

[4] To support the Government of Rwanda in the implementation of the COVID-19 National Preparedness and Response Plan the World Bank has prepared a US$14.25 million COVID-19 Emergency Response Project. In addition, as part of its support to Rwanda's anti-crisis resource mobilization, the World Bank prepared and delivered a supplemental DPO based on the series of Rwanda Energy DPOs in an amount of US$100 million.

[5] Participation in the DSSI, which provides a time-bound suspension of official bilateral debt service payments to IDA-eligible and least developed countries as defined by the UN, would free up additional resources in the near term.

[6] On the financing of the new Bugesera international airport project, it's assumed that the government will take on 40 percent of the total cost of US$1.3 billion as guaranteed debt under commercial loan financing conditions, while Qatar Airways will take on 60 percent as FDI.

[7] In the historical scenario, the PV of external debt-to-GDP ratio increases sharply since it assumes the recurrence of several large past external shocks (donor withdrawal, commodity prices, and drought) as well as large external imbalances, which were corrected over 2015–17, primarily through a large exchange rate adjustment, as envisaged under the PSI/SCF-supported program.

should pursue a proper debt management strategy, such as holding enough liquidity buffers for the crisis and smoothing out the debt servicing profile. A breach of the market-financing risk indicator also signals market financing pressures, which means that Rwanda may face liquidity pressures due to deteriorating market sentiment (Figure 6). Overall, while every solvency and liquidity indicator has at most only one short-lived breach under the baseline scenario, some indicators have multiple breaches under the most extreme shock, which indicates a moderate risk of debt distress. The government needs to adopt a credible fiscal consolidation path as soon as the COVID-19 crisis abates.

The evolution of public debt going forward is dominated by developments in the primary fiscal deficit and real GDP growth as in the past (Figure 3). Compared to the past five years, higher primary deficit due to the COVID-19 shock is expected to raise public debt despite higher GDP growth (Figure 4). Realism tools also illustrate that the baseline growth path is relatively conservative considering the envisaged fiscal consolidation by cutting domestically financed public investment and other expenditures. Furthermore, a granular assessment of the moderate risk rating shows that Rwanda has limited space to absorb shocks (Figure 5).

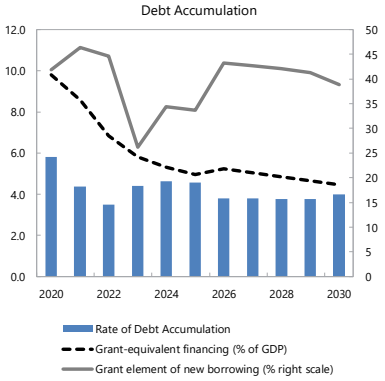
The current macroeconomic framework, which underpins this DSA, reflects currently available information. However, updates with respect to the economic impact and policy response to the COVID-19 crisis are rapidly evolving and risks to the debt outlook and sustainability are heavily tilted to the downside considering the potential of a more prolonged and deeper pandemic shock.

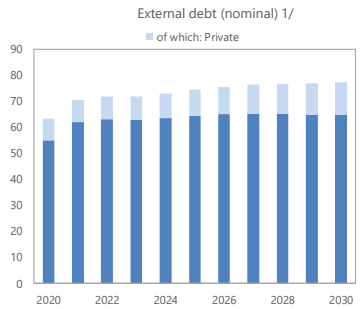
Table 1. Rwanda: External Debt Sustainability Framework, Baseline Scenario, 2017–40

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	**45.8**	**49.4**	**53.9**	**63.1**	**70.4**	**71.8**	**71.8**	**72.8**	**74.3**	**77.2**	**76.5**	**32.7**	**73.3**
of which: public and publicly guaranteed (PPG)	**37.4**	**41.1**	**45.6**	**55.0**	**61.9**	**63.0**	**62.8**	**63.4**	**64.3**	**64.7**	**56.6**	**27.3**	**63.2**
Change in external debt	3.3	3.6	4.5	9.2	7.2	1.4	0.0	1.0	1.4	0.4	-0.3		
Identified net debt-creating flows	**2.3**	**3.1**	**3.5**	**14.5**	**4.2**	**1.8**	**0.0**	**1.0**	**0.6**	**-1.5**	**-2.2**	**5.5**	**1.6**
Non-interest current account deficit	**6.2**	**6.4**	**7.7**	**15.3**	**9.0**	**8.5**	**7.4**	**7.5**	**6.6**	**5.8**	**4.6**	**8.7**	**7.6**
Deficit in balance of goods and services	10.7	11.4	11.8	16.2	12.6	12.7	7.5	8.4	7.5	6.3	5.1	**15.1**	**8.9**
Exports	21.4	21.2	21.3	13.1	21.2	22.6	26.8	25.4	25.6	27.7	32.5		
Imports	32.1	32.6	33.1	29.3	33.8	35.2	34.2	33.8	33.0	34.1	37.6		
Net current transfers (negative = inflow)	-6.4	-6.8	-5.8	-2.8	-5.4	-6.1	-3.2	-3.5	-3.5	-3.2	-3.0	**-7.7**	**-3.7**
of which: official	-6.5	-6.5	-5.5	-4.9	-5.3	-4.9	-3.3	-3.1	-2.8	-1.9	-0.8		
Other current account flows (negative = net inflow)	2.0	1.8	1.7	2.0	1.8	1.9	3.1	2.6	2.6	2.6	2.4	**1.4**	**2.5**
Net FDI (negative = inflow)	**-2.7**	**-2.9**	**-3.3**	**-1.2**	**-2.4**	**-3.1**	**-3.2**	**-3.4**	**-3.6**	**-4.0**	**-4.6**	**-2.7**	**-3.3**
Endogenous debt dynamics 2/	**-1.2**	**-0.3**	**-0.9**	**0.3**	**-2.4**	**-3.7**	**-4.2**	**-3.1**	**-2.4**	**-3.2**	**-2.3**		
Contribution from nominal interest rate	1.2	1.5	1.5	1.4	1.5	1.5	1.4	1.6	1.7	1.9	2.3		
Contribution from real GDP growth	-1.6	-3.8	-4.4	-1.0	-3.9	-5.2	-5.6	-4.7	-4.1	-5.0	-4.6		
Contribution from price and exchange rate changes	-0.8	2.0	2.0	...	...	...	...	...	...	...	...		
Residual 3/	**1.0**	**0.4**	**1.0**	**-5.3**	**3.1**	**-0.3**	**0.1**	**0.0**	**0.8**	**1.9**	**1.9**	**-1.6**	**0.5**
of which: exceptional financing	0.0	0.0	0.0	-0.2	-0.3	-0.1	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**29.3**	**34.0**	**37.7**	**38.2**	**38.5**	**39.5**	**40.8**	**41.5**	**37.5**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**137.1**	**259.3**	**177.9**	**169.2**	**143.9**	**155.7**	**159.4**	**149.7**	**115.4**		
PPG debt service-to-exports ratio	**6.5**	**7.8**	**7.5**	**22.6**	**12.0**	**11.2**	**20.3**	**9.2**	**8.7**	**9.1**	**13.2**		
PPG debt service-to-revenue ratio	**7.8**	**8.7**	**8.2**	**19.2**	**15.3**	**14.9**	**31.0**	**13.0**	**12.0**	**12.8**	**18.6**		
Gross external financing need (Billion of U.S. dollars)	0.6	0.6	0.8	2.0	1.1	1.1	1.4	1.1	1.1	1.5	4.2		
Key macroeconomic assumptions													
Real GDP growth (in percent)	4.0	8.6	9.4	2.0	6.3	8.0	8.6	7.1	6.1	7.2	6.5	**7.2**	**6.5**
GDP deflator in US dollar terms (change in percent)	1.9	-4.2	-3.9	1.0	-4.2	0.0	1.9	1.9	1.9	2.0	2.0	**-1.1**	**1.1**
Effective interest rate (percent) 4/	3.0	3.3	3.2	2.7	2.5	2.3	2.2	2.4	2.5	2.7	3.3	**3.3**	**2.5**
Growth of exports of G&S (US dollar terms, in percent)	29.1	3.1	5.7	-36.6	64.5	14.8	31.2	3.5	9.0	10.5	11.1	**14.5**	**12.7**
Growth of imports of G&S (US dollar terms, in percent)	-4.3	5.9	6.7	-8.9	17.5	12.5	7.5	7.7	5.8	10.3	10.4	**8.8**	**8.2**
Grant element of new public sector borrowing (in percent)	...	...	...	41.8	46.4	44.6	26.1	34.4	33.7	38.9	28.7	**...**	**39.6**
Government revenues (excluding grants, in percent of GDP)	17.9	19.0	19.5	15.5	16.5	17.0	17.5	18.0	18.5	19.7	23.1	**16.4**	**18.2**
Aid flows (in Billion of US dollars) 5/	0.4	0.5	0.4	1.1	1.1	0.9	0.8	0.8	0.8	1.2	1.8		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	9.8	8.6	6.8	5.8	5.3	5.0	4.5	2.8	**...**	**6.0**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	57.6	62.6	63.2	43.2	53.3	52.5	51.7	36.9	**...**	**55.6**
Nominal GDP (Billion of US dollars)	9	10	10	10	11	11	13	14	15	23	54		
Nominal dollar GDP growth	5.9	4.1	5.1	3.0	1.8	7.9	10.7	9.2	8.1	9.3	8.6	**6.0**	**7.7**
Memorandum items:													
PV of external debt 7/	...	...	37.5	42.2	46.2	46.9	47.5	48.9	50.7	54.1	57.3		
In percent of exports	...	...	175.9	321.2	217.9	208.0	177.5	192.8	198.2	194.9	176.6		
Total external debt service-to-exports ratio	12.5	15.4	15.1	35.1	19.7	18.4	26.4	15.9	15.7	17.2	24.2		
PV of PPG external debt (in Billion of US dollars)			3.0	3.6	4.0	4.4	4.9	5.5	6.1	9.5	20.2		
(PVt-PVt-1)/GDPt-1 (in percent)				5.8	4.4	3.5	4.4	4.6	4.6	4.0	2.5		
Non-interest current account deficit that stabilizes debt ratio	2.9	2.8	3.2	6.1	1.7	7.1	7.3	6.5	5.1	5.3	4.9		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

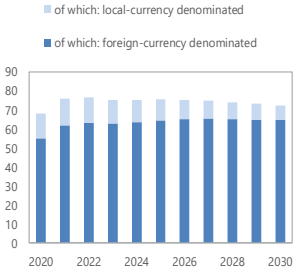
Table 2. Rwanda: Public Sector Debt Sustainability Framework, Baseline Scenario, 2017–40

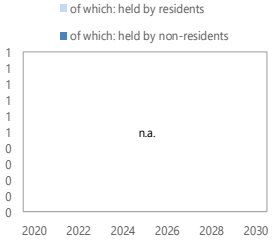
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**48.7**	**52.4**	**58.5**	**68.1**	**75.7**	**76.3**	**75.3**	**75.2**	**75.5**	**72.3**	**68.2**	**35.6**	**74.1**
of which: external debt	**37.4**	**41.1**	**45.6**	**55.0**	**61.9**	**63.0**	**62.8**	**63.4**	**64.3**	**64.7**	**56.6**	**27.3**	**63.2**
Change in public sector debt	4.3	3.7	6.1	9.6	7.6	0.6	-1.1	0.0	0.3	-0.7	-0.5		
Identified debt-creating flows	**0.8**	**2.8**	**5.5**	**9.5**	**7.8**	**0.9**	**-0.8**	**0.1**	**0.3**	**-0.6**	**-0.6**	**2.0**	**1.4**
Primary deficit	**3.6**	**3.5**	**6.8**	**10.2**	**8.0**	**5.2**	**5.1**	**5.1**	**4.5**	**4.2**	**3.3**	**3.1**	**5.3**
Revenue and grants	22.6	23.8	23.6	20.1	20.7	20.6	20.6	20.9	21.2	21.5	23.9	**23.4**	**21.0**
of which: grants	4.7	4.8	4.2	4.6	4.1	3.6	3.1	2.9	2.7	1.8	0.9		
Primary (noninterest) expenditure	26.2	27.3	30.5	30.3	28.7	25.8	25.8	26.0	25.7	25.7	27.2	**26.5**	**26.4**
Automatic debt dynamics	**-2.6**	**-0.6**	**-1.4**	**-0.7**	**-0.2**	**-4.2**	**-5.9**	**-4.7**	**-4.0**	**-4.8**	**-3.8**		
Contribution from interest rate/growth differential	-1.6	-2.7	-3.8	-1.5	-3.4	-5.4	-5.9	-4.7	-4.0	-4.8	-3.9		
of which: contribution from average real interest rate	0.1	1.1	0.7	-0.3	0.6	0.2	0.1	0.3	0.3	0.1	0.3		
of which: contribution from real GDP growth	-1.7	-3.8	-4.5	-1.1	-4.0	-5.6	-6.0	-5.0	-4.3	-4.9	-4.2		
Contribution from real exchange rate depreciation	-1.0	2.1	2.5	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**-0.1**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.3**	**-0.2**	**0.0**	**0.0**	**-0.1**	**-0.1**
Privatization receipts (negative)	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (use of earmarked fund)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-0.3	-0.2	0.0	0.0		
Residual	**3.4**	**0.9**	**0.6**	**0.9**	**3.0**	**0.8**	**-0.3**	**-0.2**	**0.0**	**-0.1**	**0.1**	**2.0**	**0.3**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**42.8**	**48.2**	**52.5**	**52.5**	**51.9**	**52.4**	**53.0**	**50.2**	**50.0**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**181.1**	**239.7**	**254.1**	**254.1**	**251.8**	**250.5**	**250.0**	**233.4**	**209.0**		
Debt service-to-revenue and grants ratio 3/	**26.5**	**29.3**	**27.0**	**40.2**	**41.4**	**42.5**	**54.5**	**39.7**	**37.1**	**31.3**	**34.7**		
Gross financing need 4/	8.4	9.0	13.2	18.3	16.5	13.9	16.4	13.1	12.2	10.9	11.6		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	4.0	8.6	9.4	2.0	6.3	8.0	8.6	7.1	6.1	7.2	6.5	**7.2**	**6.5**
Average nominal interest rate on external debt (in percent)	2.0	2.3	2.3	1.7	1.7	1.6	1.4	1.7	1.7	1.7	1.8	**1.9**	**1.7**
Average real interest rate on domestic debt (in percent)	-1.9	6.9	5.2	-0.7	7.0	3.0	3.3	3.3	3.5	3.6	4.2	**2.4**	**3.4**
Real exchange rate depreciation (in percent, + indicates depreciation)	-3.2	6.0	6.5	...	...	...	...	...	...	...	...	**3.0**	**...**
Inflation rate (GDP deflator, in percent)	7.6	-0.8	0.4	6.5	1.0	5.0	5.0	5.0	5.0	5.0	5.0	**3.5**	**4.8**
Growth of real primary spending (deflated by GDP deflator, in percent)	7.1	13.2	22.2	1.4	0.5	-2.8	8.4	8.2	4.7	7.5	7.6	**10.4**	**4.9**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-0.7	-0.2	0.8	0.6	0.4	4.5	6.2	5.2	4.3	4.9	3.8	**0.0**	**4.1**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security and extra budgetary funds, central bank, government-guaranteed debt, non-guaranteed SOE debt. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

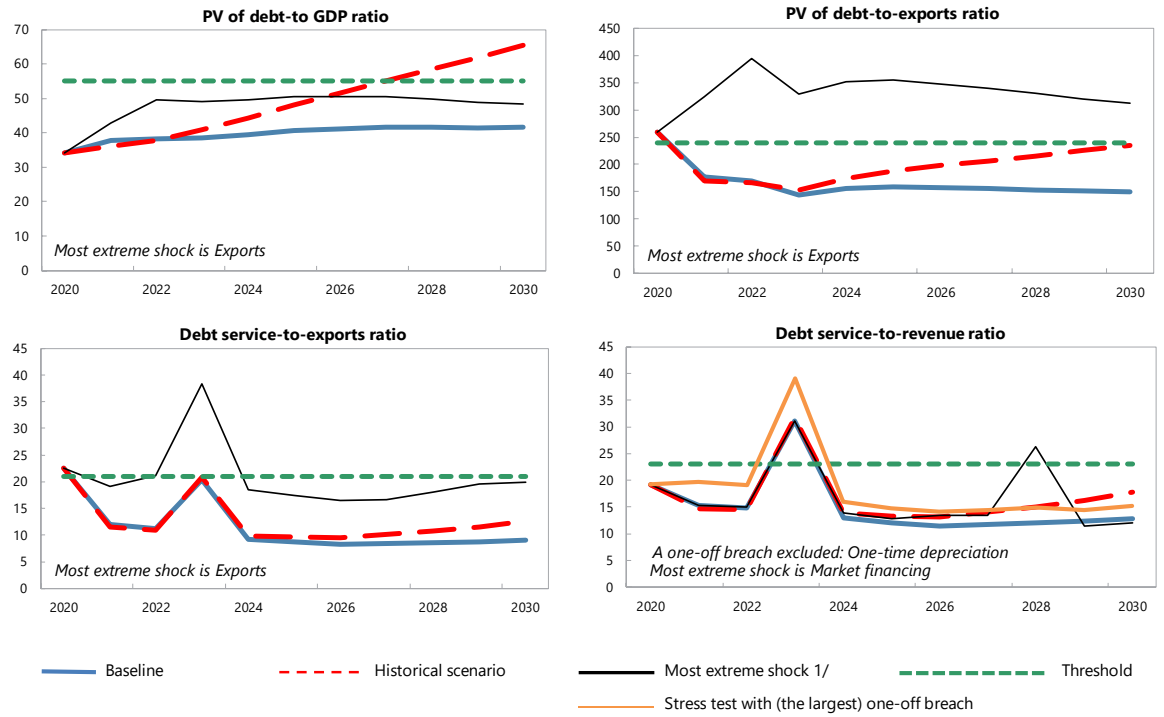
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Rwanda: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2020–30 1/ 2/



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	n.a.	n.a.
Market Financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.0%	2.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	29	29
Avg. grace period	6	6

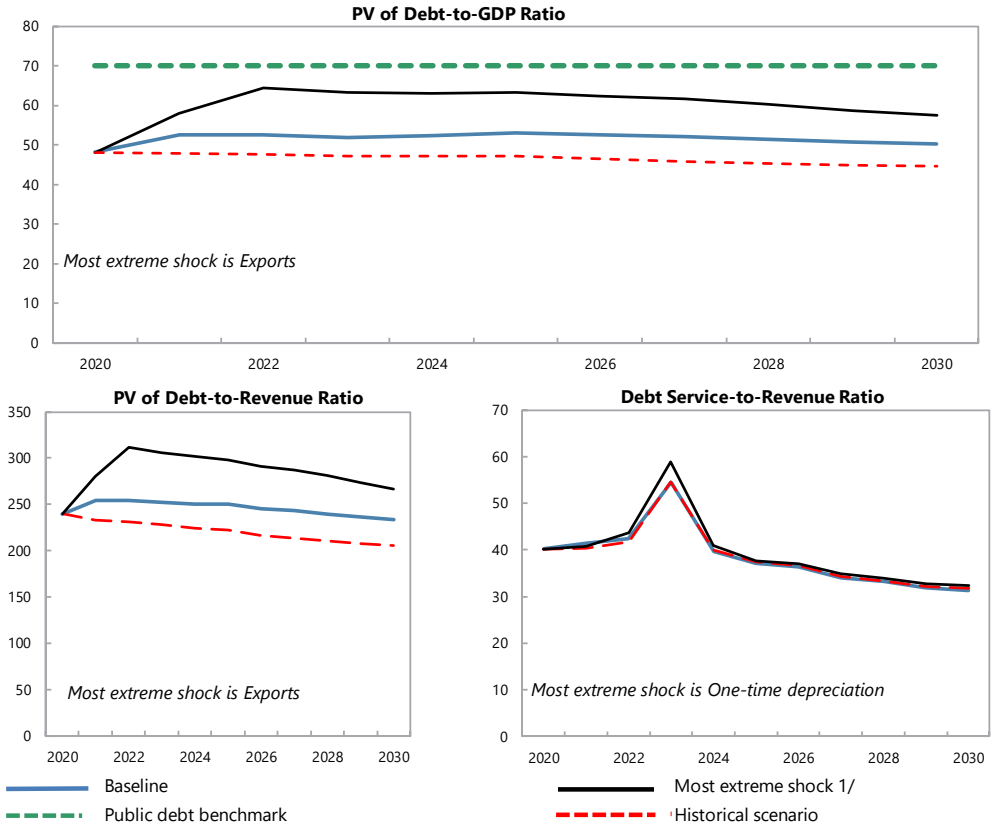
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Rwanda: Indicators of Public Debt Under Alternative Scenarios, 2020–30 1/



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	60%	60%
Domestic medium and long-term	14%	14%
Domestic short-term	26%	26%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.0%	2.0%
Avg. maturity (incl. grace period)	29	29
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	5.6%	5.6%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	1%	1.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Rwanda: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020–30

(In percent)

	Projections										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	**34.0**	**37.7**	**38.2**	**38.5**	**39.5**	**40.8**	**41.2**	**41.5**	**41.5**	**41.4**	**41.5**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	34.0	36.1	37.7	40.9	44.2	48.1	51.5	55.0	58.4	61.8	65.3
B. Bound Tests											
B1. Real GDP growth	34.0	38.9	40.5	40.8	41.9	43.2	43.7	44.0	44.0	43.9	44.0
B2. Primary balance	34.0	38.5	41.0	41.3	42.3	43.6	44.0	44.2	44.1	43.9	43.9
B3. Exports	34.0	42.9	49.5	49.1	49.6	50.5	50.5	50.5	49.9	48.9	48.3
B4. Other flows 2/	34.0	39.6	41.8	41.9	42.8	43.9	44.2	44.4	44.2	43.9	43.7
B6. One-time 30 percent nominal depreciation	34.0	48.3	44.5	45.2	46.6	48.4	49.1	49.7	49.8	50.0	50.5
B6. Combination of B1-B5	34.0	45.7	46.2	46.2	47.0	48.2	48.4	48.6	48.2	47.7	47.4
C. Tailored Tests											
C1. Combined contingent liabilities	34.0	40.5	41.6	41.9	43.1	44.4	44.7	45.0	44.9	44.6	44.6
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	34.0	42.4	42.9	43.3	44.5	46.0	46.5	46.9	45.9	45.7	45.8
Threshold	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**
PV of debt-to-exports ratio											
Baseline	**259.3**	**177.9**	**169.2**	**143.9**	**155.7**	**159.4**	**158.1**	**155.7**	**153.1**	**150.9**	**149.7**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	259.3	170.2	167.1	152.7	174.2	188.0	197.7	206.2	215.4	225.3	235.5
B. Bound Tests											
B1. Real GDP growth	259.3	177.9	169.2	143.9	155.7	159.4	158.1	155.7	153.1	150.9	149.7
B2. Primary balance	259.3	181.4	181.6	154.5	166.9	170.6	168.8	165.9	162.7	159.8	158.1
B3. Exports	259.3	324.0	393.8	329.4	351.0	354.2	347.9	339.7	330.0	319.9	312.5
B4. Other flows 2/	259.3	186.6	185.4	156.6	168.6	171.7	169.6	166.6	163.1	159.7	157.7
B6. One-time 30 percent nominal depreciation	259.3	177.9	154.0	131.8	143.6	147.8	147.2	145.5	143.5	142.3	142.1
B6. Combination of B1-B5	259.3	311.2	192.7	259.8	279.0	283.5	279.8	274.4	267.5	261.4	257.5
C. Tailored Tests											
C1. Combined contingent liabilities	259.3	190.9	184.1	156.4	169.8	173.5	171.7	168.8	165.6	162.7	160.9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	259.3	177.9	169.3	144.0	156.3	160.3	159.0	156.6	150.8	148.3	147.1
Threshold	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**
Debt service-to-exports ratio											
Baseline	**22.6**	**12.0**	**11.2**	**20.3**	**9.2**	**8.7**	**8.3**	**8.4**	**8.6**	**8.7**	**9.1**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	22.6	11.5	11.0	20.8	9.9	9.6	9.5	10.1	10.7	11.5	12.6
B. Bound Tests											
B1. Real GDP growth	22.6	12.0	11.2	20.3	9.2	8.7	8.3	8.4	8.6	8.7	9.1
B2. Primary balance	22.6	12.0	11.5	20.8	9.7	9.2	8.7	8.8	9.1	9.3	9.6
B3. Exports	22.6	19.1	21.2	38.3	18.5	17.4	16.5	16.6	18.0	19.7	19.9
B4. Other flows 2/	22.6	12.0	11.4	20.7	9.6	9.0	8.6	8.7	9.1	9.5	9.7
B6. One-time 30 percent nominal depreciation	22.6	12.0	11.2	20.0	8.9	8.4	8.0	8.1	8.3	8.1	8.5
B6. Combination of B1-B5	22.6	18.1	19.1	33.3	15.6	14.7	14.0	14.1	15.6	15.6	16.0
C. Tailored Tests											
C1. Combined contingent liabilities	22.6	12.0	11.6	20.6	9.6	9.0	8.6	8.7	8.9	9.0	9.4
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	22.6	12.0	11.3	20.4	9.9	9.3	9.7	9.7	18.8	8.2	8.6
Threshold	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**
Debt service-to-revenue ratio											
Baseline	**19.2**	**15.3**	**14.9**	**31.0**	**13.0**	**12.0**	**11.4**	**11.7**	**12.0**	**12.3**	**12.8**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	19.2	14.7	14.5	31.8	13.9	13.2	13.1	14.0	15.0	16.2	17.7
B. Bound Tests											
B1. Real GDP growth	19.2	15.8	15.8	32.9	13.7	12.7	12.1	12.4	12.7	13.0	13.5
B2. Primary balance	19.2	15.3	15.3	31.8	13.7	12.7	12.0	12.3	12.7	13.1	13.6
B3. Exports	19.2	15.3	15.7	32.6	14.5	13.4	12.7	12.9	14.1	15.4	15.6
B4. Other flows 2/	19.2	15.3	15.2	31.6	13.5	12.4	11.8	12.1	12.7	13.3	13.7
B6. One-time 30 percent nominal depreciation	19.2	19.6	19.1	39.1	16.0	14.8	14.1	14.5	14.9	14.5	15.2
B6. Combination of B1-B5	19.2	16.1	16.8	33.8	14.6	13.5	12.8	13.1	14.5	14.6	15.0
C. Tailored Tests											
C1. Combined contingent liabilities	19.2	15.3	15.3	31.5	13.5	12.5	11.9	12.2	12.5	12.7	13.2
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	19.2	15.3	15.0	31.2	13.9	12.8	13.4	13.4	26.3	11.5	12.0
Threshold	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Includes official and private transfers and FDI.

Table 4. Rwanda: Sensitivity Analysis for Key Indicators of Public Debt, 2020–30

(In percent)

	Projections										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	**48.2**	**52.5**	**52.5**	**51.9**	**52.4**	**53.0**	**52.5**	**52.1**	**51.4**	**50.7**	**50.2**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	48	48	48	47	47	47	46	46	45	45	45
B. Bound Tests											
B1. Real GDP growth	48	54	57	57	58	59	59	60	59	59	59
B2. Primary balance	48	54	56	56	56	56	56	55	54	53	53
B3. Exports	48	58	64	63	63	63	62	62	60	59	57
B4. Other flows 2/	48	54	56	55	56	56	56	55	54	53	53
B6. One-time 30 percent nominal depreciation	48	58	56	53	51	51	49	47	45	44	42
B6. Combination of B1-B5	48	51	53	52	52	53	52	51	51	50	49
C. Tailored Tests											
C1. Combined contingent liabilities	48	58	57	57	57	57	56	56	55	54	53
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	48	52	52	52	53	53	53	52	51	50	50
Public debt benchmark	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**
PV of Debt-to-Revenue Ratio											
Baseline	**239.7**	**254.1**	**254.1**	**251.8**	**250.5**	**250.0**	**245.2**	**242.9**	**239.7**	**236.2**	**233.4**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	240	233	231	229	225	222	216	213	210	208	206
B. Bound Tests											
B1. Real GDP growth	240	262	272	273	275	278	276	276	276	275	274
B2. Primary balance	240	261	274	270	267	265	259	256	252	248	245
B3. Exports	240	280	312	306	302	298	291	287	281	273	267
B4. Other flows 2/	240	263	272	269	266	265	260	257	253	248	244
B6. One-time 30 percent nominal depreciation	240	285	273	258	248	241	230	222	213	205	197
B6. Combination of B1-B5	240	251	258	254	251	250	244	240	236	231	228
C. Tailored Tests											
C1. Combined contingent liabilities	240	281	278	274	271	269	263	260	256	252	248
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	240	254	254	252	251	251	246	244	237	233	230
Debt Service-to-Revenue Ratio											
Baseline	**40.2**	**41.4**	**42.5**	**54.5**	**39.7**	**37.1**	**36.4**	**34.0**	**33.2**	**32.0**	**31.3**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	40	40	42	55	40	37	37	34	33	32	32
B. Bound Tests											
B1. Real GDP growth	40	42	45	59	44	42	41	39	39	38	37
B2. Primary balance	40	41	45	58	42	40	38	35	35	33	33
B3. Exports	40	41	43	56	41	38	38	35	35	35	34
B4. Other flows 2/	40	41	43	55	40	38	37	34	34	33	32
B6. One-time 30 percent nominal depreciation	40	41	44	59	41	38	37	35	34	33	32
B6. Combination of B1-B5	40	40	43	57	41	38	37	34	33	32	31
C. Tailored Tests											
C1. Combined contingent liabilities	40	41	52	58	45	40	38	35	34	33	32
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	40	41	43	55	40	38	38	36	46	31	31

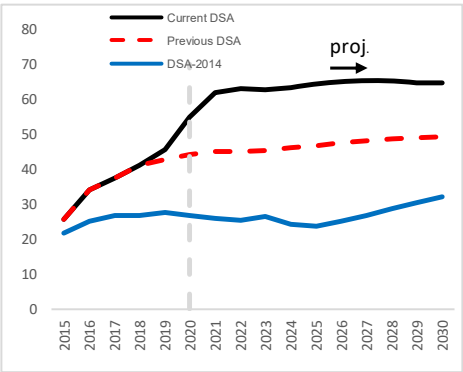
Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

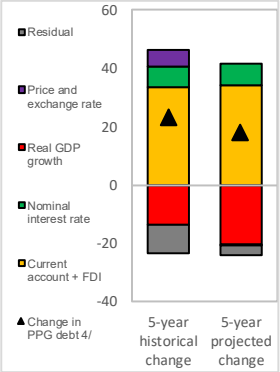
2/ Includes official and private transfers and FDI.

Figure 3. Rwanda: Drivers of Debt Dynamics—Baseline Scenario[1/]

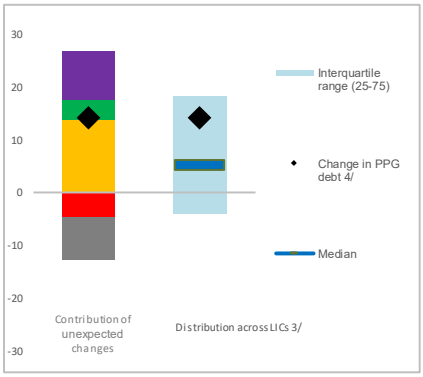
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

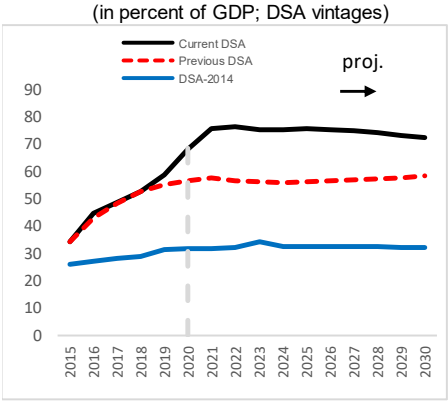


Unexpected Changes in Debt 2/
(past 5 years, percent of GDP)

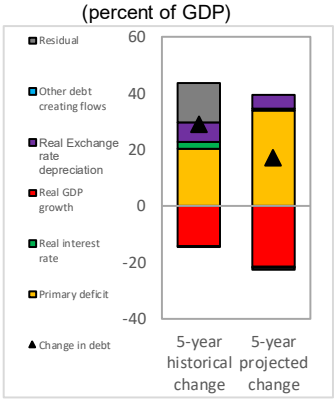


Public debt

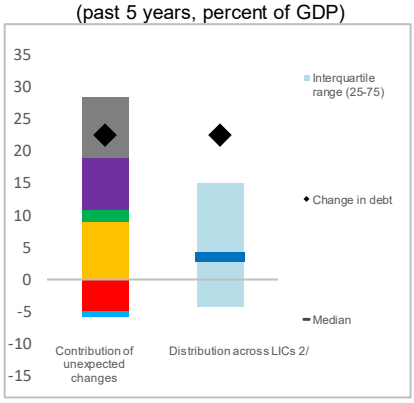
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Compared to 2014 DSA scaling up scenario

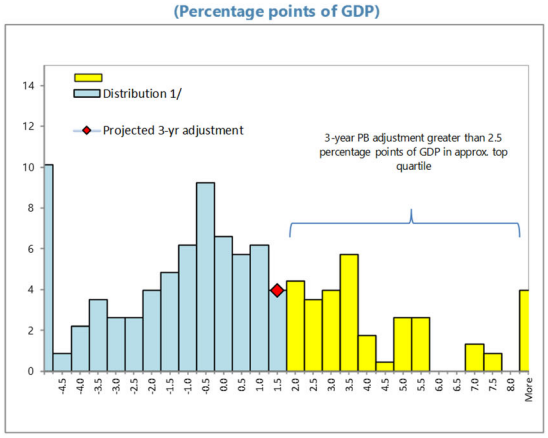
2/ Difference between anticipated and actual contributions on debt ratios.

3/ Distribution across LICs for which LIC DSAs were produced.

4/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

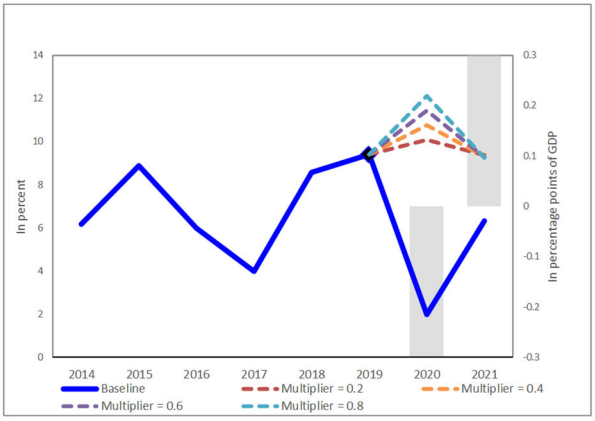
Figure 4. Rwanda: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



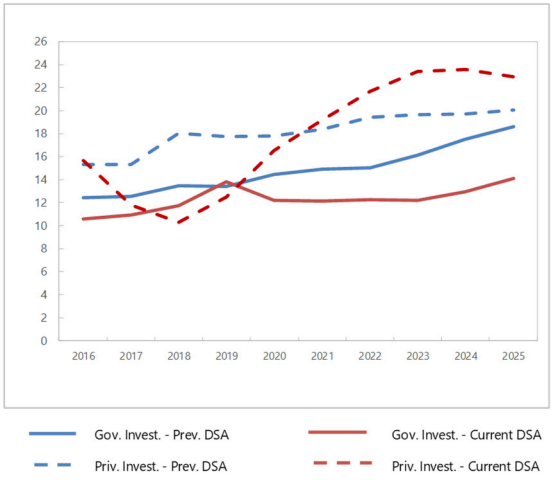
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates 1/
(% of GDP)



1/ Private investment includes the Bugesera airport project.

Contribution to Real GDP growth
(percent, 5-year average)

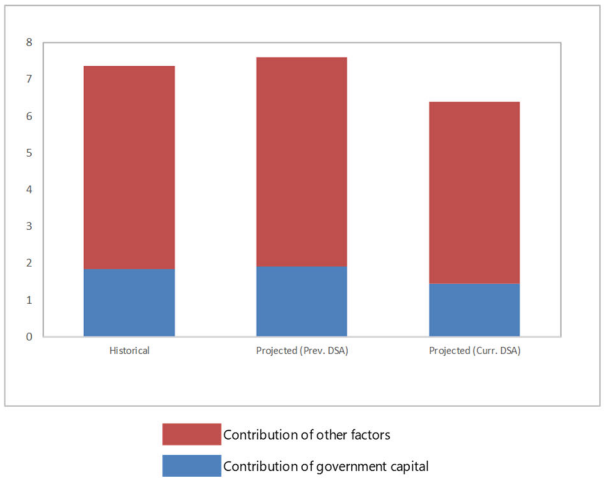


Figure 5. Rwanda: Qualification of the Moderate Category, 2020–2030[1/]

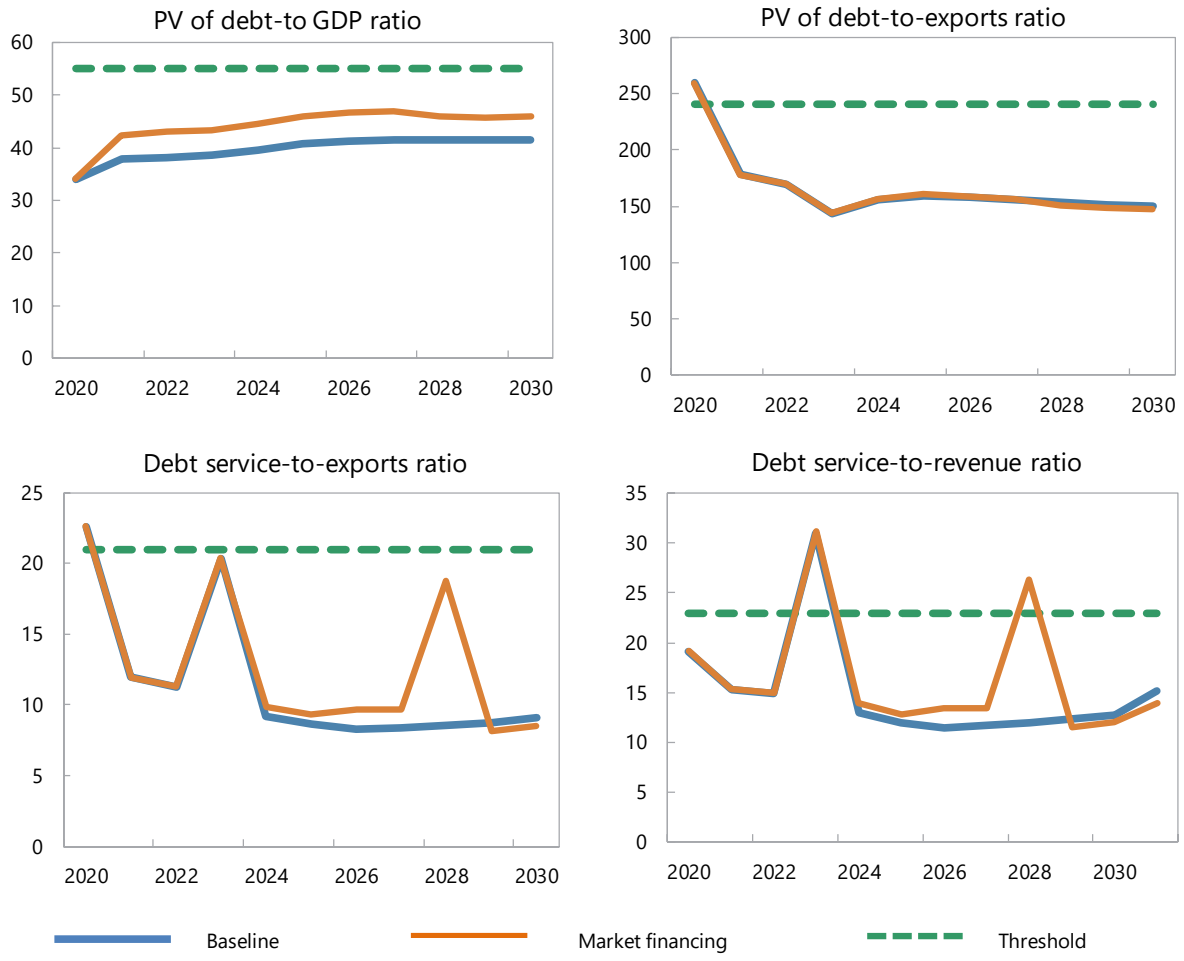
Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

Figure 6. Rwanda: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	18	n.a.
Breach of benchmark	Yes	n.a.
Potential heightened liquidity needs	High	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.